GLOBAL FUEL PRICES, TAXATION AND SUBSIDY ANALYSIS

A Tableau Dashboard-Based Study of Global Fuel Markets
2020–2026

Project focus

An analysis of petrol, diesel and LPG prices, regional taxation patterns, government subsidies, income-group comparisons, diesel price volatility and price forecasting.

Dataset coverage: 27,468 records · 84 countries · 7 global regions

Seven-page analytical report prepared from the project description and the supplied Tableau dashboard reference. No images are included in this document.

1. Executive Summary

Fuel is an essential input for households, industry, transportation and international trade. Differences in fuel prices can influence the cost of living, business competitiveness and the movement of goods. Governments also affect the final price paid by consumers through taxes, subsidies and other policy measures.

This project uses Tableau to examine global fuel-price patterns from 2020 to 2026. The dataset contains 27,468 records covering 84 countries across 7 global regions. The dashboard brings together nine visualizations and organizes the analysis around three stakeholder perspectives: an Energy Policy Consultant, a Global Supply Chain Manager and an Energy Economics Researcher.

The dashboard combines geographical comparison, grouped bars, price comparisons, a time-series forecast, a country ranking, a volatility heat map, income-group charts and a price heat map. Together, these views make complex relationships easier to explore and communicate.

Objectives of the project

• Compare fuel-related indicators across countries and regions.
• Explore how taxation and subsidies vary across regional categories.
• Examine diesel price trends, country-level differences and volatility.
• Compare taxation, subsidy distributions and fuel prices across income groups.
• Present the findings in an interactive and understandable Tableau dashboard.

The central questions are whether fuel indicators differ by region, how tax and subsidy categories are distributed, which countries show higher average diesel prices, how volatility changes over time, and how fuel measures compare across income groups.

2. Project Scope and Methodology

The analysis is structured around three fuel types—petrol, diesel and LPG—and examines them through geography, time, subsidy category and national income group. The selected period, 2020–2026, supports historical comparison and the presentation of forecasted diesel-price movement.

Dataset overview

Dataset dimension	Project coverage
Records	27,468
Countries	84
Global regions	7
Time period	2020–2026
Fuel types	Petrol, diesel and LPG
Visualizations	9

Analytical approach

Descriptive analysis: The map, regional tax and subsidy comparison, price comparison, country ranking and income-group charts summarize patterns in the data.

Time-series analysis: The diesel forecast chart displays historical movement and a projected continuation beyond the observed period. Forecasts are estimates, not guaranteed outcomes.

Comparative analysis: Heat maps and income-group charts compare several categories at once, making patterns and exceptions easier to identify.

The dashboard includes filters and legends that allow users to narrow the view by year, region, subsidy level, income level and selected measures. These controls support focused exploration. The charts can reveal associations and differences, but they do not independently prove that a particular tax or subsidy caused a specific price outcome.

3. Scenario One: Energy Policy Consultant

Stakeholder: Dr. Sarah Mitchell

The first stakeholder is an Energy Policy Consultant interested in understanding how governments use taxation and subsidies and how these policy choices relate to fuel prices. For this stakeholder, the dashboard emphasizes geographical differences and comparisons of tax, subsidy and price indicators.

Visualization 1 — Global Fuel Map

The map provides a country-level geographical view of the selected fuel-related measure. Its value is that it places national indicators in a regional context. A consultant can use it to identify clusters, contrasts and countries that may deserve closer investigation. It is a starting point for asking why neighbouring or economically similar countries may display different values.

Visualization 2 — Tax Rate by Region and Subsidy Level

This grouped bar chart compares average tax percentages across regions and subsidy categories. It allows the user to examine whether tax levels appear different among countries grouped by low, medium or high subsidy levels. The regional breakdown matters because a global average could conceal substantial differences between areas.

Visualization 3 — Price vs Subsidy

The price comparison chart places average fuel price values alongside subsidy categories and regions. It helps the consultant explore whether higher subsidy categories tend to coincide with lower observed prices, and whether the pattern is consistent across regions. Differences between regions may reflect market structure, tax policy, supply costs or other factors that require further study.

Policy relevance

Together, these views can support discussions about affordability, revenue collection and targeted assistance. They help identify questions for deeper analysis, such as whether subsidies are associated with lower consumer prices, whether tax burdens differ by region, and whether policy outcomes vary between countries. A full policy evaluation should also consider household income, public finances, import dependence, exchange rates and refining capacity.

4. Scenario Two: Global Supply Chain Manager

Stakeholder: Marcus Thompson

The second stakeholder is a Global Supply Chain Manager. Fuel is a major operating input for road freight and many other logistics activities. Changes in diesel prices can affect budgets, route economics and procurement decisions.

Visualization 4 — Diesel Price Forecast

The line chart presents historical diesel-price movement and a forecast beyond the historical period. The historical line helps identify broad changes over time, while the forecast gives an estimated direction for future prices. A manager can use this information when preparing fuel budgets, reviewing supplier contracts and planning procurement. Forecast uncertainty should be considered, especially when unexpected geopolitical, economic or supply events occur.

Visualization 5 — Top 10 Countries

This horizontal bar chart ranks the countries with the highest average diesel prices. It makes it easier to compare countries and identify locations where diesel may represent a comparatively high cost. For supply chain planning, the chart can flag markets that need closer cost monitoring. Country-level prices should also be interpreted alongside local purchasing power, taxes, transport distances and actual business routes.

Visualization 6 — Diesel Price Volatility by Region and Year

The volatility heat map compares diesel-price volatility across regions and years. Its colour scale helps identify periods and regions with relatively higher or lower volatility. High volatility can make operating costs less predictable, while lower volatility may support more stable budgeting. The manager can use this view to identify where contingency budgets, flexible contracts or more frequent price reviews may be useful.

Operational implications

These visualizations support three practical activities: **forecasting** expected fuel expenditure, **benchmarking** costs across markets and **managing risk** where price movements are less stable. They do not replace route-level cost models, but they provide useful global context for operational decisions.

5. Scenario Three: Energy Economics Researcher

Stakeholder: Professor Elena Rodriguez

The third stakeholder is an Energy Economics Researcher who wants to examine how fuel taxation, subsidy structures and prices compare across low-, middle- and high-income economies. This scenario focuses on differences between income groups rather than only geographical regions.

Visualization 7 — Tax by Income

The bar chart compares average fuel tax percentages across low-, middle- and high-income groups. It provides a direct view of differences in average tax levels. The researcher can explore whether income categories are associated with different approaches to fuel taxation. These comparisons may help generate hypotheses about fiscal capacity, energy policy priorities and the role of fuel taxes in government revenue.

Visualization 8 — Subsidy by Income

The 100% stacked bar chart shows how subsidy categories are distributed within each income group. Because each bar represents 100%, the chart is useful for comparing proportions rather than absolute country counts. It can reveal whether low, medium or high subsidy categories make up a larger share of a particular income group and frame questions about fuel affordability across economies.

Visualization 9 — Price by Income

The price heat map compares average fuel price indicators across income groups. Colour intensity helps identify relative differences across the displayed measures. This can support research into the relationship between income classification and fuel pricing. Income alone should not be assumed to explain price differences: taxes, subsidies, exchange rates, production costs, import dependence and market regulation may also contribute.

Research relevance

The income-group visualizations offer a comparative foundation for studying energy affordability and policy design. They can help researchers identify patterns worth testing with more detailed statistical methods, such as regression analysis or country-level case studies. The charts are descriptive and should be treated as evidence of observed differences, not proof of causation.

6. Overall Insights, Recommendations and Conclusion

Overall insights

The dashboard demonstrates the value of examining fuel markets from multiple perspectives. Regional comparisons highlight geographical variation; the diesel forecast and volatility heat map introduce a time dimension; and the income-group charts provide an economic comparison. No single chart explains the global fuel market on its own, but the nine visualizations together create a broader analytical picture.

Recommendations

• **For policymakers:** use regional tax and subsidy comparisons to identify policy differences and investigate whether support reaches intended consumers.
• **For supply chain managers:** combine forecasts and volatility indicators with route-specific fuel consumption, supplier quotations and scenario-based budgets.
• **For researchers:** use income-group comparisons to develop testable hypotheses and investigate the influence of other variables on fuel prices.

Limitations

This report is based on the project description and the dashboard information provided in the conversation. It does not include the complete underlying dataset, data dictionary, calculation definitions or forecast settings. Therefore, it explains the purpose and analytical use of each visualization without claiming statistical significance or causal effects. Before using the results for formal policy or business decisions, the underlying data should be validated and the definitions of tax, subsidy, price and volatility should be documented.

Conclusion

In conclusion, the project uses Tableau to transform 27,468 records across 84 countries and 7 global regions into nine accessible visualizations. The dashboard addresses the needs of an Energy Policy Consultant, a Global Supply Chain Manager and an Energy Economics Researcher. It demonstrates how data visualization can support comparison, reveal patterns and guide further investigation into fuel prices, taxation and subsidies.

Global fuel pricing is complex. It reflects the interaction of government policy, regional conditions, market forces and economic circumstances. A well-designed dashboard helps decision-makers explore these factors together and ask better questions about the global energy market.

Thank you.